



The perils of working with big data, and a SMALL checklist you can use to recognize them



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Abstract The use of big data to help explain fluctuations in the broader economy and key business performance indicators is now so commonplace that in some instances it has even begun to rival more traditional measures. Big data sources can very often provide advantages when compared with these more traditional data sources, but with these advantages also come potential pitfalls. We lay out a checklist called SMALL that we have developed in order to help interested parties as they navigate the big data minefield. Based on a set of five questions, the SMALL checklist should help users of big data draw justifiable conclusions and avoid making mistakes in matters of interpretation. To demonstrate, we provide several case studies that demonstrate the subtle nuances of several of these new big data sets and show how the problems they face often closely relate to age-old concerns that more traditional data sources are also forced to tackle.

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1. The promise and challenges of big data

More and more people are appealing to the use of big data to explain fluctuations in economic activity and to track key business performance

indicators. In recent years, greater numbers than ever before have used the large and novel data sets of private and public companies ([The Economist, 2020a](#)) to measure the health and economic consequences of the COVID-19 epidemic. In this article, we take a closer look at the efficacy of big data in this context and provide a novel checklist to guide consumers of these new big data sets.

The granularity and high frequency of big data make it useful for tracking business conditions in

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real time (Chetty et al., 2020). These features are both advantages when compared with more traditional sources—such as official government statistics—that usually are reported with a substantial lag and often offer only limited geographic and demographic detail (Abraham et al., 2020). However, with these advantages also come some potential pitfalls. More traditional data sources have the benefit of being time-tested, and their vagaries are understood well and documented publicly. This is rarely the case for big data, at least in its infancy, and often it can lead to mistakes (Lazer et al., 2014).

In what follows, we lay out a checklist that we have developed in order to help navigate the big data minefield. Referred to as SMALL in an obvious play on words, it is meant to draw a distinction between what big data purports to be and what it can be instead if not used carefully. Drawing from a series of case studies that we describe below, we argue that the problems big data faces are often closely related to age-old concerns that official statistics—and the agencies charged with producing them—have also been forced to tackle. Once these problems are identified with our SMALL checklist, solutions often exist to address them, but where they do not, a healthy dose of skepticism is warranted.

We are by no means the first to go down this path; several of our concerns here can also be found in the analyses of others. Previous work has shown that with the new opportunities big data provides (e.g., Alharthi et al., 2017; Earley, 2015; Gupta et al., 2020; Pham & Stack, 2018) comes a new set of challenges ranging from data quality, privacy considerations, and the required infrastructure to manage big data (Lee, 2017; Nagle et al., 2020) to the abilities of a firm to monetize the data it collects and to produce meaningful analytics from it (Walker, 2015).

The focus of this article—and of the SMALL checklist—is instead on the bigger picture of using big data to track business conditions in real time. For this reason, the case studies we describe share a significant time series component. It is here that we hope that our collection of cautionary tales will be most useful. As both business organizations and government statistical agencies are beginning to make inroads to the use of big data (Daas et al., 2015), reexamining its properties has never been more important. We show that this introspection can be made easier by asking five questions, one for each element of the SMALL acronym. Asking these questions should lead to more-accurate inferences and help to avoid making mistakes with potential policy or business implications.

2. The SMALL checklist

The SMALL checklist, shown in Figure 1, poses five questions to guide the use of big data. It is from these questions that we derive the SMALL acronym:

1. How are the data **S**ampled? (Are the data representative in all dimensions?)
2. How are the data **M**easured? (Is it consistently constructed over time?)
3. How are the data **A**ssembled? (Are statistical methods used to alter the data?)
4. Do the data exhibit reporting **L**ags? (Are there reporting delays?)
5. Are the data a **L**eading indicator? (Are they predictive of business conditions?)

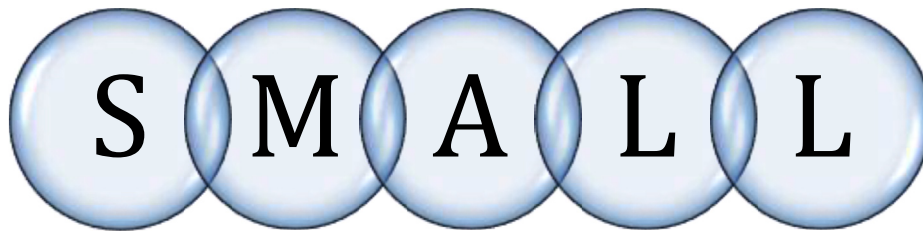
At this point, readers may be asking themselves: “Why should I bother asking these questions as long as I go in knowing that the data has limitations?” And in some ways, that sentiment is correct. The problem, however, often comes in defining what those limitations are in the first place, why they present themselves, and what might be a possible solution. The SMALL checklist makes this process more transparent by making it easier to identify and convey the limitations of big data sources, while also pointing to possible solutions.

The five questions of the SMALL checklist—and the answers to them—are crucial for any successful empirical analysis with big data because of what have come to be known as the five Vs (Lee, 2017): volume, veracity, variety, velocity, and value. In fact, each of the five V dimensions of big data motivates one of the five questions within the SMALL checklist (see Column 2, Figure 1).

2.1. Sampled

Asking how the data are sampled is crucial when the volume of the data being used is large. Volume, being a multidimensional measure of size, can be large for many reasons. Does the large volume come from an increased granularity in the type of transactions for a single firm? Or does it come from having a particular type of transaction for many customers? Or does it come from a mix of both? Where volume comes from has substantial implications on what analysis is appropriate for the

Figure 1. The SMALL checklist



<i>Checklist Question</i>	<i>5V-Big Data Dimension</i>	<i>Main Finding</i>
1. How are the data Sampled?	Volume	Size is not a substitute for coverage in sampling
2. How are the data Measured?	Veracity	Accuracy begins with proper measurement
3. How are the data Assembled?	Variety	When assembling the data, choose the right tool for the job
4. Do the data exhibit reporting Lags?	Velocity	Duration is not the sole criterion for judging reporting lags
5. Are the data a Leading indicator?	Value	Leading information requires finding the right variation in the data

data. The main takeaway from the first question of the SMALL checklist is this: Highly detailed data from an individual firm might be incredibly revealing of its operations but not at all relevant for understanding the market, industry, or economy on a broader scale.

2.2. Measured

Asking how the data are measured is directly related to uncovering the veracity of the data being used. Mismeasurement, changes in measurement procedures, and gaps in measurement all represent just a few of the examples of the sorts of issues that can plague a big data source. Why might these issues be more prevalent in a big data setting? With the sheer scale of measurements taking place in many big data sets, the opportunities for these measurement issues to arise are more abundant, but they need not completely prohibit using the data. By knowing where and why the measurement issues arise, it should be possible to accommodate them into an analysis of the data's scope and appropriate uses.

2.3. Assembled

Asking how the data are assembled acknowledges the reality that in many big data applications, the variety of data is significant. This variety can lead to big data sets containing information at different frequencies (i.e., some variables are available each day, while other variables are available each hour), different locations, and substantially different interpretations (e.g., total versus per capita, real versus nominal, stock variable versus flow variable). Typically, the big data collector (e.g., Google) also assembles these potentially varied measures of information using statistical methods to make them comparable (e.g., Google Trends). Sometimes the appropriate method to achieve such a comparison is obvious—like scaling sales across different locations in per capita terms. Other times the appropriate method is less obvious, or worse, depends on the question to be answered or the analysis desired. In this case, an apples-to-apples comparison is only as good as the statistical method used to assemble it.

2.4. Lags

Asking whether the data exhibit any reporting lags underscores the role of velocity in big data. Many big data sources provide access to data that are constantly updating, reflecting near real-time information, and these updates are occurring at increasingly higher frequencies. This trend would seem to indicate that any concerns about reporting lags are limited, though this could not be further from the truth. As the velocity of big data increases, maintaining careful attention to how reporting lags change over time, differ across units, and relate to the type of activity being tracked has now become all the more relevant for big data users.

2.5. Leading

Asking how the data may act as a leading indicator for business conditions addresses directly one potentially broad source of value for big data. While the ability to produce vital information on key business performance indicators in a more timely fashion is certainly valuable, predictions about where business conditions are heading are likely to be even more so. In many cases, capturing this value from any big data source requires identifying the highest quality and most representative data to help answer your question.

3. Using the SMALL checklist

With this brief overview of the SMALL checklist serving as the backdrop, we now turn to illustrating its use through a series of five case studies.

3.1. How are the data sampled?

Unlike the methods used to construct more-traditional data sets, big data are not commonly structured in a way that would allow them to address broader questions (e.g., “Is economic activity for the U.S. declining?”). Economists tend to call this an issue of *external validity*. While a data set may be perfectly fine for addressing the characteristics of a firm’s customer base, it need not be representative of a firm’s industry or of the broader economy.

This is not a critique of big data collection so much as a reflection of the potentially different aims of the data collector and the data user. To illustrate, consider the construction of the consumer price index (CPI) by the U.S. Bureau of Labor

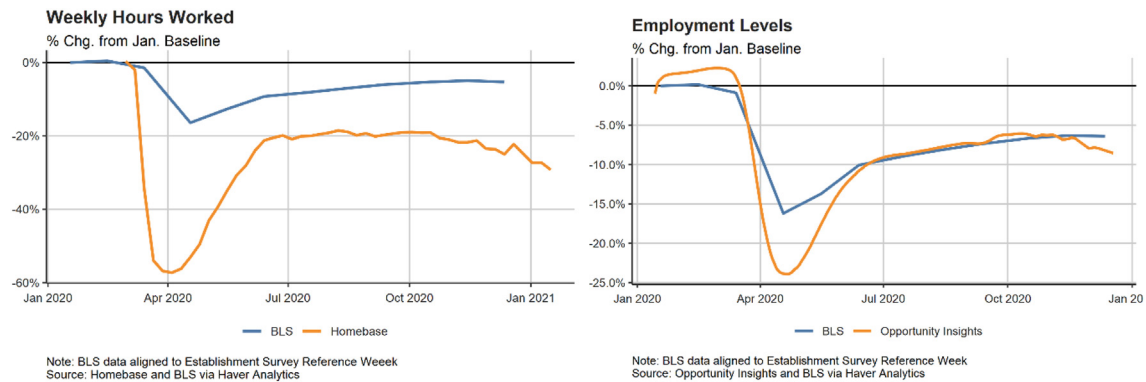
Statistics (BLS). As [Friedman et al. \(2019\)](#) discuss, the BLS recently began experimenting with using big data in the CPI. Rather than send people out to price items in stores, they reached out to large retailers instead for this information since they were already collecting it. On the surface, this seems like a very natural application of leveraging the capabilities of big data, but it was not as successful as one might expect. When large retailers record price data, they are primarily interested in keeping track of what will help them to maximize profits. This is not the goal of the BLS, which is instead primarily interested in facilitating the computation of the CPI (e.g., accounting for quality-adjustment concerns).

Big data sources can sometimes be nothing more than *convenience samples* of a specific type of customer or business activity that is easily tracked, typically as the means for some other business function. In contrast, users of big data are often instead interested in *representative samples* of a desired population. For this reason, we prefer to think of big data as a sample drawn from a potentially broader universe of interest, but one that at the same time is also not likely to be drawn at random since it is generated by a conscious decision-making process by an individual to engage in some sort of behavior (e.g., purchasing a product, browsing a website).

Typically, a goal of the big data collector is to be as unstructured as possible in order to cast a wide net for these types of actions. Therefore, using big data in other contexts requires that a bridge of sorts be built to map the activity that is measured to the structure of the more general economic concept of interest. In some cases, big data collectors/providers build this bridge for the user. An example here is the PriceStats index of online prices discussed in [Cavallo and Rigobon \(2016\)](#) and [Cavallo \(2017\)](#) that closely tracks the CPI.

How this is done, though, can bring about other concerns in our SMALL checklist. While private firms often sell their data, these firms are not statistical agencies charged with making all of their methods publicly available. There is no guarantee that the proprietary methods used to create the data are fully explained and documented, and even when they are, they can still be subject to vigorous debate and interpretation.

The use of cell-phone data is a good example. One of the premier sources of information used to measure the potential for the spread of COVID-19, this type of data has been highly studied (e.g., [Atkinson et al., 2020](#); [Vanhoof et al., 2018](#)). Still,

Figure 2. Weekly average hours worked and employment relative to similar measures from BLS

its representativeness has been called into question by Coston et al. (2020). By linking one popular source of this data for the U.S. with voter rolls, the researchers found that older and nonwhite voters are less likely to be captured. Given the higher mortality rates among these populations, such a result is potentially of concern to public health officials charged with tracking the spread of the virus. Below, we provide similar examples for other dimensions of business conditions during the pandemic based on big data sources of labor market information.

3.1.1. Application: The COVID-19 labor market

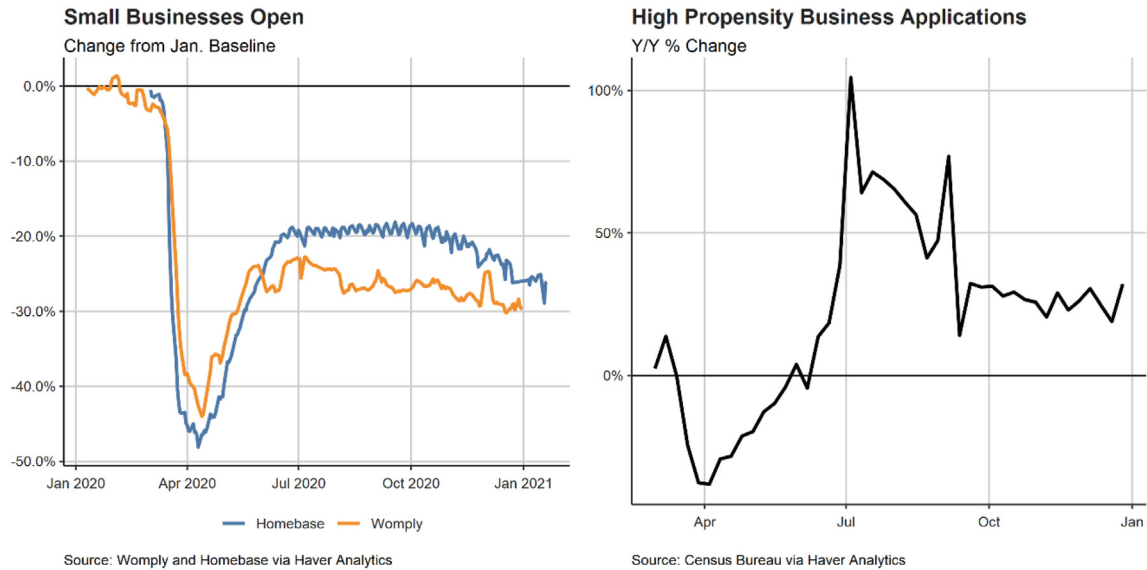
Although the BLS's Employment Situation report provides among the timeliest of labor market statistics, there has still been considerable demand during the pandemic for even more up-to-date labor market data, which has driven customers to new big data sources. To illustrate two of these newly popular sources, Figure 2 shows measures of weekly average hours worked and employment relative to similar measures from the BLS. In both cases, we have aligned the timing of the BLS survey data to the big data source to facilitate comparison, but it is worth noting that both big data sources are available at even higher frequencies. To highlight changes over time, all the measures in the figure have been indexed against a similar week in January of 2020 and show deviations from this level through December 2020.

The measure of hours worked comes from Homebase, a firm that provides time scheduling software. While both the Homebase and BLS measures of hours worked capture a steep decline in the spring followed by a recovery and stabilization to a lower level by summer, their magnitudes are quite different: The decline in the

Homebase data is about three times as large as the decline in the BLS data. This discrepancy is largely driven by the fact that the Homebase firms differ from the firms tracked in the BLS survey in two important ways: First, the Homebase firms tend to be concentrated in industries such as restaurants, retail, and personal services that were most affected by the public health measures imposed to mitigate the spread of the virus. Second, the Homebase sample skews toward smaller businesses because the firms that use the Homebase product are primarily individually owned or operator-managed. These underlying differences illustrate why the Homebase data should not be expected to be representative of the labor market as a whole.

While it is difficult to look at the left panel of Figure 2 and come away with anything but a sense of awe at the impact of the pandemic on hours, it is also quite apparent that not everyone was affected equally by it. This is an important aspect of the SMALL checklist's first question. Representativeness can have stark implications for public policy when the incidence of a policy or event is experienced unequally. Big data can help to identify those effects, but we must first know what it is we are and are not capturing. For instance, those workers who were able to successfully transition to remote work arrangements during the pandemic likely had a very different experience than the workers captured by Homebase. For some of them, hours worked may have even increased with reductions in commuting time and lower at-home productivity (Barrero et al., 2020).

The right panel of Figure 2 provides a second example. Here, we compare changes in the level of employment from the BLS survey with a measure constructed by the Opportunity Insights (OI) lab with big data from the firms Paychex, Earnin,

Figure 3. The extensive margin of business operations during the pandemic

Intuit, and Kronos. The OI series is constructed in a sophisticated way that attempts to account for both firm size and industry in order to better match the overall composition of employers. This matching process can be viewed as an example of the type of bridge we discussed above linking the big data source to the general concept of interest. The declines and subsequent increases in employment during the pandemic in the OI data are more in line with the BLS survey. In this case, the bridge built was a good first step toward addressing the SMALL checklist's first question.

3.2. How are the data measured?

While problems to do with representativeness are usually restricted to observations within a specified time period, a whole host of additional problems arise when the researcher is also interested in making comparisons across time periods. Here, problems often revolve around *internal validity* and concern the ability of a data set to accurately capture what it purports to describe. For instance, a data set that shows foot-traffic patterns at geographic places of interest may be used to judge the state of retail activity. But what happens when the way this data is collected differs across two points in time? If the researcher's aim is to draw comparisons across time in order to identify changes in demand, concerns of this nature cloud all aspects of the data's use even for its intended purpose.

As an example, consider what would happen if the underlying number of sensors in such a data set were to double unbeknownst to the researcher. This would likely cause the number of hits to increase quite substantially, but does it represent a true increase in activity? The answer is likely to be no, and more often than not, issues of this nature prevent building the kind of bridge that is necessary to connect big data to other, more traditional data sources, including official statistics. More traditional data sources are also not immune to these considerations, but they typically have developed time-tested methods to address them and that make comparisons over time feasible.

One example is how firm entry and exit affects the measurement of productivity (Olley & Pakes, 1996). The underlying complication in this case hinges on the fact that firms typically enter or exit a data set for reasons that are far from random, and this is usually only partially revealing of the firm's underlying productivity—the very thing that we are trying to measure. Most firms fail (i.e., exit) within the first year (Abbring & Campbell, 2005), and for reasons typically tied to inferior productivity. At the same time, a firm's (de novo) entry into a particular market is usually construed as the culmination of a series of particularly good events that have made the proprietor optimistic of the firm's capabilities. In either case, the firm might differ from the typical firm in the industry. Thus, researchers hoping to assess how productivity has evolved over time are forced to think

carefully about how to handle the classic selection bias problem inherent in data sets with entry and exit.

One insight from this work is the discovery that firm entry and exit can have different roles than firm expansion and contraction in driving the evolution of key variables. For instance, aggregate productivity in an industry—say telecommunications—could increase not only because the firms in the industry individually become more productive but also because the firms entering the industry are more productive than the firms exiting it. Finally, even without any entry or exit or improvement in firm-level productivity, aggregate productivity might increase if production is allocated toward more-productive firms and away from less productive firms. That is to say, the more productive firms expand and the less productive firms contract. As we show in the next application, these measurement issues that are commonplace in more traditional data sources are just as—if not more—relevant for big data.

3.2.1. Application: Measuring exit and entry in real time

The Homebase data have been crucial for obtaining up-to-date information about the labor market during the COVID-19 pandemic (Bartik et al., 2020). However, consumers of this data must also consider how it is measured over time. Comparing the Homebase data with that of a competing firm called Kronos provides an illustrative example. Homebase and Kronos take different approaches to constructing their panels of firms. Homebase uses a fixed panel. This means that if a business starts using their service, they are *not* added to the panel, while if a business stops using it (which is distinct from having zero recorded working hours), it is dropped from the panel. In contrast, Kronos measures the total number of shift punches and payrolls without accounting for changes in the sample of firms that use its software.

These two methods provide different insights into changes in employment and hours worked. By using a fixed panel, Homebase captures changes along what economists refer to as the *intensive margin*, or the amount of labor that existing firms demand. The Kronos data instead captures both changes along the intensive margin as well as changes along the *extensive margin*, or the number of firms that are demanding labor. Both margins are of interest in their own right. The intensive margin may be especially interesting in the context of the COVID-19 pandemic, as it may help in understanding how well businesses are weathering the pandemic. At the same time, the

developments along the extensive margin (firm entry/exit) are an important component of aggregate labor market conditions.

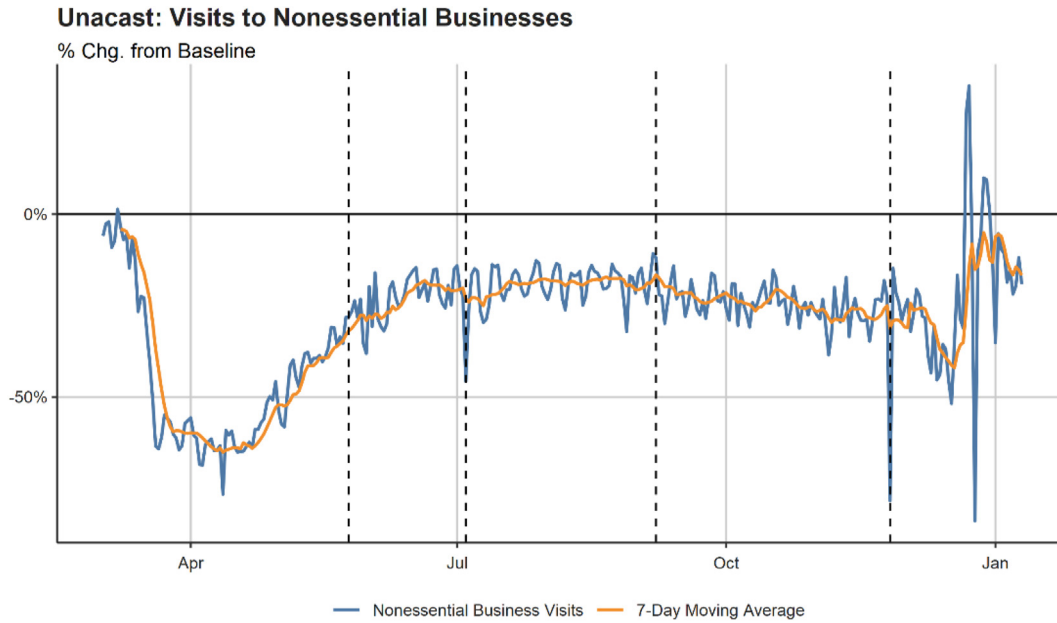
Figure 3 shows two data series that together illustrate the extensive margin of business operations during the pandemic. The left panel of the figure contains data from Homebase and Womply, a credit card payment aggregator, comparing the number of businesses open to a January reference period. Like the Homebase data, Womply also uses a fixed panel of firms defined as those that are regularly open from January through February. Because both big data sources compare a similar set of firms at two different points in time, they capture the extensive margin of business closures.

The Homebase and Womply data, however, cannot easily tell us whether these closures are temporary or permanent, nor even how they compare to other recessions given their short history. To answer those questions, other big data sources are required. For example, Yelp (2020), the business review website, noted in September 2020 that, of businesses that were open on March 1 and had marked their business as closed, 60% indicated that they were permanently closed. Similarly, Crane et al. (2021) combined several big data sets to evaluate firm exit during the pandemic from a historical perspective.

The right panel of Figure 3 instead shows new business applications that are identified by the U.S. Census Bureau as having a high propensity to become a business with a payroll. This measure of new business formation captures the extensive margin of firm entry. Through the summer months of 2020, many firms were still shut down owing to the pandemic and public health restrictions, but since the end of June 2020, there has also been a wave of new businesses to fill some of this gap. Anecdotal evidence (Guilford & Scott, 2020; The Economist, 2020b) suggests that these new businesses are largely small operations started by individuals whose prior employment was disrupted by the pandemic.

Given what we see in Figure 3, drawing conclusions about the overall labor demand of small employers during the pandemic solely on the basis of the Homebase or Womply data is probably not a good idea. In doing so, one would likely miss a considerable amount of variation along the extensive margin. The same can be said, though, for the data sources that better capture firm entry. For instance, there is a well-known lag between new business applications and actual employment, so putting too much emphasis on that data alone is also probably not a good idea if the goal is to understand current labor market trends. Instead, to

Figure 4. Cell-phone data: Number of visits to nonessential U.S. businesses during the pandemic



Source: Unacast

get the whole picture, it is better to consider all of these big data sources together with traditional sources of information on the labor market.

3.3. How are the data assembled?

Big data users must also take note of how the data are assembled—in particular, what, if any, transformations were used to construct them and whether any effort has been made to adjust them for seasonal fluctuations. Many transformations used with traditional data sources are not as reliable to use on big data. Often, the best way of capturing changes in big data sources is to focus on year-over-year or reference period comparisons. This is a very attractive way to answer the question, “How different are these data from normal?”

Using comparisons to a reference period, however, can still create issues with holidays and other semiregular but atypical events of a seasonal nature. Seasonality is a topic that will be familiar to many close followers of indicators that have a time series component to them. Seasonal fluctuations are the dominant source of variation for industries like retail grocery (Butters et al., 2020; Chevalier et al., 2003), hotels and air travel (Butters, 2020), and construction (Geremex & Gourio, 2018). Even growth in real gross domestic product (GDP) experiences a seasonal trough in the first quarter each year that dwarfs the size of most recessions (Barsky & Miron, 1989).

The breadth and depth of seasonal effects often requires an adjustment be made in order to isolate the variation of interest instead, such as adjusting for the business cycle of GDP. Of course, to the extent that the adjustment fails to fully capture seasonality, it can lead to improper inferences. This is why analysts obsess over seasonal considerations in the daily flow of information to track business conditions in real time, and why it is something that statistical agencies address in painstaking detail (U.S. Census Bureau, 2017).

Because big data are typically measured at a very high frequency, the established methods used by statistical agencies to correct for seasonality are not feasible. Alternative methods do exist (Cleveland & Scott, 2007; McElroy, 2017; McElroy et al., 2018), but they tend to be considerably more difficult to implement. Instead, what is often done is to appeal to transformations that can be thought of as simple seasonal adjustments. An example is the use of year-over-year comparisons in which days or weeks are matched across years. Even this clever trick, however, often still leaves behind traces of residual seasonality. Below, we highlight a few salient examples using cell-phone data.

3.3.1. Application: Transformations and seasonality in cell-phone data

Despite their obvious advantages, applying fixed-reference-period comparisons to daily big data

time series is not a panacea. For instance, year-over-year comparisons can still face issues when tied to particular days of the week or to holidays that move around from year to year (e.g., Easter, Thanksgiving). Using a fixed period as a baseline comparison can also create issues when the data demonstrate strong seasonality, although in both cases this can often be partially mitigated by using a 7-day moving average of the data.

To demonstrate, [Figure 4](#) shows cell-phone data from Unacast for the number of visits to nonessential businesses in the U.S. during the pandemic. The figure nicely illustrates both points. The large negative spikes correspond to Memorial Day, July 4th, Labor Day, and Thanksgiving. Without knowing their cause, it would be easy to misinterpret them, but it is also clear that using a 7-day moving average is a little more robust to these concerns.

In addition to the holiday effects noted for the Unacast data, there are very likely seasonal patterns in cell-phone mobility data of this nature as well (e.g., fewer visits to workplaces during the summer, fewer recreation trips to beaches and parks in the winter) that cannot be disentangled from the underlying changes in mobility when the data are constructed in this manner. Specifically, the use of a fixed base period does not allow for the fact that the amount of mobility almost certainly varies over the course of the year and by business type.

During the pandemic, social distancing restrictions have been prevalent. This may not matter too much in terms of capturing the overall decline in mobility with data of this nature, but it will affect the precision of the estimates made over time as the seasons change from spring/summer to fall/winter. This is quite evident in the Unacast data, for instance, around the December/January holiday period. Furthermore, this is not something that a 7-day moving average can correct. To establish a seasonal norm requires at least one additional year of data to use for comparison.

3.4. Do the data exhibit reporting lags?

One of the most attractive features of many of the new big data sources that have been used to track the economy during the pandemic is that they are available more rapidly than more traditional sources (e.g., annual reports, official statistics). For instance, while the lag time on many economic indicators, such as employment or GDP, is on the order of weeks to months, many big data sources are available in near real time. That said, often there are still structural hurdles that prevent the data from being truly useful in real time.

Understanding these publication delays should always be a first-order concern.

Publication delays typically reflect the lag time between when the activity of interest occurs and when the data are first made available to analyze; this could be anywhere between one to two days to one to two weeks, depending on the activity and the data source. But the potentially more important delay tends to occur between the first release of the data for a given time period and when it can be considered final. For some big data sources, especially those based on cell-phone location, the first release of the data can be considered definitive. However, for an important subset of big data sources, most notably credit and debit card spending data, there is an appreciable delay before the data can be considered final. In this case, it can be necessary to allow several extra days for observations to filter into the data set; otherwise, analysts are at risk of detecting a spurious decline at the end of the time series.

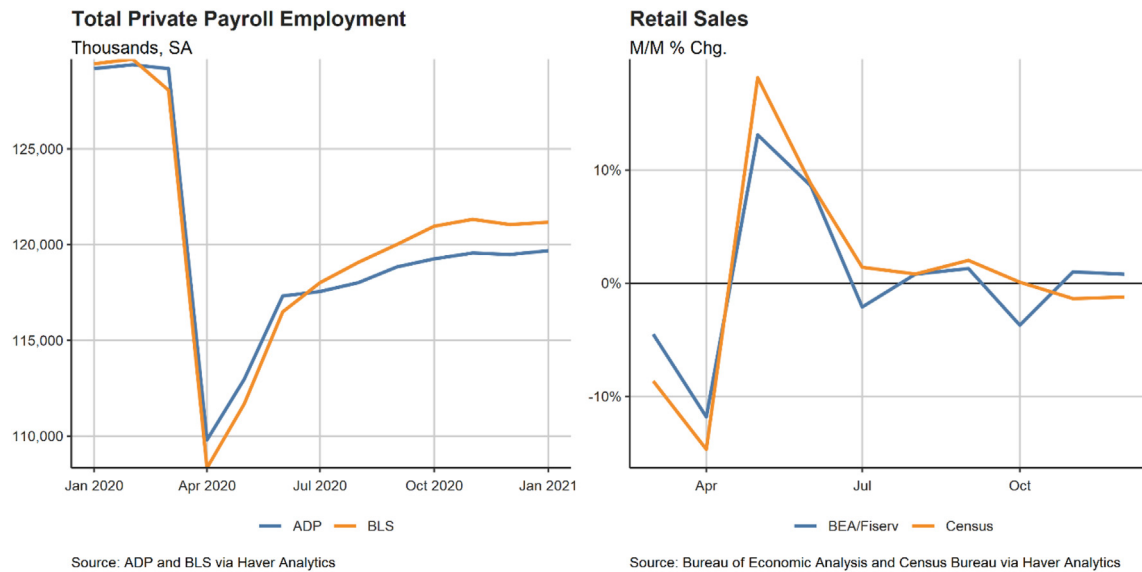
Before using a big data source, it is critical to understand whether any structural processing delay exists, how long it is, and how this type of reporting delay could affect the conclusions drawn from the data. An example of this with traditional data is the use of citation counts in the study of research and development ([Azoulay et al., 2019](#); [Bryan & Ozcan, 2020](#)). Citation counts are used as a proxy for the dissemination of information and for the value of such activity. But censoring of recently produced research can be problematic. In other words, the relevant (even relative) citation count of a patent filed in 2001 will differ greatly from a comparable patent filed in 2020 for reasons unrelated to either patents' usefulness. While research designs do exist to overcome this fact, this setting still underscores the other types of channels through which reporting lags may exist.

3.4.1. Application: Processing delays in credit and debit card transactions

For credit and debit card transaction data, reporting lags are a consequence of processing delays in payment networks (i.e., the time it takes from a transaction clearing at the point of sale to its settlement between financial institutions). The settlement lag is typically 1 to 3 business days, making it important to take this lag into account when analyzing this data. That said, lags can differ greatly across different parts of the payment network, and much work is already underway to reduce them (e.g., [Herbst-Murphy, 2013](#)).

The bigger picture that we want to emphasize with this example is that knowledge of the institutional arrangements that govern big data

Figure 5. ADP data relative to BLS estimates of total private payroll employment throughout the pandemic and nascent recovery



collection is crucial to understanding and properly using it. This issue comes up, for instance, when comparing retail sales to data published by the U.S. Bureau of Economic Analysis (BEA), which uses credit and debit card transactions from Fiserv, a large payment intermediary (Aladangady et al., 2019). The use of this data by the BEA is a good example of how reporting delays can be made manageable when properly understood. Retail sales are used by the BEA as source information for U.S. GDP. Since GDP is a quarterly statistic, a delay of a few days is not of major concern for GDP data. Like many big data sources, determining the value of the credit and debit card transaction data instead comes down to its predictive ability, as we detail next.

3.5. Are the data a leading indicator?

For many big data users, a data set's utility is likely to lie largely in how predictive it is of business conditions. One version of this is the process known as "nowcasting," which has increasingly been adapted to include big data sources. Here, the lead time between the big data source and the key variable of interest is critical. A good forecast, or nowcast, that is available for weeks or a month ahead of time is much more valuable than one of the same accuracy that only comes out a day or two in advance. Additionally, the degree of leading information—as measured by how predictive the data are and by the ease with which such

predictions can be generated—is also an important factor in the data's ultimate value.

Similarly, given their relative novelty, big data sources generally do not have long histories. This complicates truly testing their ability to generate reliable out-of-sample predictions in real time, as the power of any such test will be naturally limited. With not enough data points to firmly establish the statistical significance of any predictive relationship, it falls to the economic significance of the argument to carry the day. Here, we come full circle because economic significance is closely tied to the way the data are collected. In order to establish economic significance, the researcher must have a firm understanding of what the data are measuring and how the data set was generated. In other words, the data and conclusions drawn from it are likely to be discounted by others if by design it is a "black box." Transparency is, therefore, almost itself a prerequisite for the use of big data in nowcasting.

3.5.1. Application: Leading signals from payroll and card processors

Several big data sources would seem to already fit the bill when it comes to being accurate predictors of business conditions, and, undoubtedly, more will be developed and tested as the need arises. One of the earliest and best-documented examples is the data collected by the payroll-processing firm ADP and published each month ahead of the BLS Employment Situation report. As Figure 5

shows, the ADP data have tracked quite closely with the BLS estimates of total private payroll employment throughout the crisis and nascent recovery—of that, there is little doubt. But to return to the point raised above, how much value is there in knowing the ADP number a mere few days before the BLS number?

The answer to that question depends on the audience, perhaps, but there could be other factors at play that still make the ADP data highly informative. The BLS data, for instance, are from a survey of establishments and are subject to revision due to lagged reporting and other miscellaneous features (including the issue of entry and exit of firms covered earlier). Since the ADP data can be predictive of these revisions (Cajner et al., 2018), their lead time is actually longer because the BLS revises its survey results only twice: once when the following month's results are released, and another time in March of each year when the previous year's results are updated to reflect incoming administrative source data. More recent work has even tried to extend the lead time of the ADP data further by constructing it at a weekly frequency instead (Cajner et al., 2020).

The COVID-19 pandemic has also led to a great deal of attention on the credit and debit card data. Previously, we noted the use of the Fiserv transaction data by the BEA. Fiserv is one of many similar such big data firms. Brave et al. (2021) combined the information from a number of these other credit and debit card data processors with retail foot traffic data collected from cell phones and daily surveys of consumer sentiment to produce a weekly index of retail trade. By combining the related information from all of these big data sources, the authors built the kind of bridge we referred to earlier that allows for a way of filtering through some of the high-frequency noise in the credit and debit card data to reveal the smoother signal that is often more relevant for predicting underlying trends in business conditions.

4. The road ahead

Big data has produced an excellent opportunity to inform business strategies and policy decisions with highly granular and timely information on business conditions. These opportunities, however, come with the need to be mindful of the obstacles those same data present. Fortunately, many of these obstacles have been faced before with traditional data sources. Through a series of case studies, we have shown that the five distinct

aspects of our novel SMALL checklist, capturing sampling design, measurement, data assembly, reporting lags, and leading/predictive ability, have important implications on the types of conclusions that can be drawn from big data and thus its ultimate value. By collecting these lessons learned from past statisticians, econometricians, and other data scientists, our SMALL checklist should help big data users to avoid the potential pitfalls that can arise.

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